Two end modes: logarithmic asymptotics and importance sampling for the peak load of a pickup-and-delivery route

Quang-Vinh Dang[1*]

[1*]British University Vietnam, Hung Yen, Vietnam.

Corresponding author(s). E-mail(s): vinh.dq4@buv.edu.vn;

Abstract

A vehicle that delivers and collects goods along a route carries, after each stop, the deliveries still to be made plus the pickups already made. We study the probability that the peak of this load exceeds a large level $\boldsymbol{c}$ when the demands are gamma variables coupled by one-factor Gaussian copulas, one for deliveries and one for pickups. We prove $\boldsymbol{\log \mathbb{P}(\text{peak} > c) = -c/\lambda^\star + O(\sqrt{c})}$, where $\boldsymbol{\lambda^\star}$ is the larger of two spectral scales, roots of secular equations for the delivery block at the start of the route and the pickup block at its end. A single mean shift at the dominant end is logarithmically efficient when the weaker end scale is below $\boldsymbol{\lambda^\star/4}$ and the dominant block has a spectral gap, and is not efficient above $\boldsymbol{\lambda^\star/4}$. The failure can be silent: when few draws are expected in the weaker end mode, the estimate has conditional mean equal to the probability minus that mode's mass, with a small sample variance. A mixture of shifts at the stage dominating points with a defensive component is logarithmically efficient under a covering condition that reduces to a two-dimensional min–max problem, with closed-form special cases. On 5 920 routes of two regional plans the condition holds, for constant weights, on every route at copula correlation $\boldsymbol{0.6}$; checks against a quadrature reference confirm the predicted underestimation of the single shift and the accuracy of the mixture down to $\boldsymbol{10^{-6}}$.

Keywords: rare-event simulation, importance sampling, logarithmic efficiency, large deviations, Gaussian copula, vehicle routing

MSC Classification: 65C05 , 60F10 , 90B06

1 Introduction

A vehicle leaves a depot with the goods for L customers and, at each stop, unloads the delivery d_i and loads the pickup p_i. After stop k it carries

$$L_k = \sum_{i>k} d_i + \sum_{i\le k} p_i, \qquad k = 0, \dots, L,$$

and it overflows a capacity c if the peak $F = \max_k L_k$ exceeds c. This is the load profile of the vehicle routing problem with simultaneous pickup and delivery (Dethloff 2001). When the demands are random, planning models control $\mathbb{P}(F > c)$ at moderate levels (Gendreau et al. 2016; Gounaris et al. 2013; Dinh et al. 2018; Ghosal and Wiesemann 2020); medical, cold-chain and dangerous-goods distribution ask for levels between 10^{-3} and 10^{-6}, which must then be verified by simulation, and at such levels crude Monte Carlo needs $1.96^2/(\varepsilon^2 p)$ draws for relative error ε.

The probabilistic question is the tail of a maximum of dependent sums. The load process (L_k) is not a random walk: its increments $p_k - d_k$ change sign, the first term $D_k = \sum_{i>k} d_i$ decreases and the second $P_k = \sum_{i\le k} p_i$ increases along the route, and the demands within each type are correlated. We take gamma marginals and a one-factor Gaussian copula for each type, $d_i = G_{d,i}^{-1}(\Phi(Z_{d,i}))$ with $Z_{d,i} = \sqrt{\rho}\, W_d + \sqrt{1-\rho}\, E_{d,i}$, and likewise for pickups with an independent factor W_p. The gamma quantile of a normal score grows like $\theta z^2/2$, so each L_k is, to leading order, a positive part of a quadratic form in the Gaussian inputs, and its tail is governed by the largest eigenvalue of a matrix $\Theta^{1/2} C \Theta^{1/2}$ built from the gamma scales and the copula correlation (the same structure appears in delta–gamma approximations of portfolio losses, Glasserman et al. 2000).

Contributions.

(i) Theorem 1: for every stage, $\log \mathbb{P}(L_k > c) = -c/\lambda_k + O(\sqrt{c})$ (with an $O(\log c)$ upper error), and $\log \mathbb{P}(F > c) = -c/\lambda^\star + O(\sqrt{c})$, where λ_k is the larger of the scales of the stage's delivery and pickup blocks, each the largest root of an explicit secular equation, and $\lambda^\star = \max_k \lambda_k$ is attained, when $\rho > 0$, only at the two ends of the route (Corollary 1). On the logarithmic scale a route overflows in one of two *end modes*: all deliveries on board at the start, or all pickups on board at the end.

(ii) Theorem 2: let $\varrho \in (0, 1]$ be the ratio of the weaker to the stronger end scale. A single mean shift at the dominant end is not logarithmically efficient when $\varrho > 1/4$ (with a defensive component: when $\varrho > 1/2$), and it is logarithmically efficient when $\varrho \le 1/4$ and the dominant end block has a spectral gap of at least 4.

(iii) Proposition 1: in the inefficient case the second moment is inflated by a region the proposal reaches only as often as crude Monte Carlo does. On the event, of probability at least $1 - N\,\mathbb{P}(\text{weaker end event})$, that no draw reaches it, the estimate has conditional mean $p - \mathbb{P}(\text{weaker end event})$ up to a factor $1 + O(p)$; its own sample variance does not reveal the deficit. We show that this predicts the bias measured on real routes, of the single shift and of a two-step Glasserman–Li estimator.

(iv) Theorem 3: a mixture of mean shifts at the asymptotic dominating points of the stage blocks, with weights that may decay like $e^{-c\delta_m}$ and a defensive component, is

logarithmically efficient when every stage satisfies a covering inequality $\Psi_k \geq 4\lambda_k/\lambda^\star$. The constant Ψ_k is the value of a two-dimensional min–max problem in the stage's spectral ratios; Lemma 5 evaluates it in closed form in the cases that matter ($\Psi = 4$ when one block of the stage dominates the other by a factor 3 and both have a spectral gap of 3; $\Psi \geq 2\sqrt{2}$ under a gap of 2, with equality for a tie), and Corollary 3 extends the result to several routes sharing the factors.

(v) Section 7 checks the conditions on 5 920 routes of plans for two regions (Northern Vietnam and Guangdong) and compares the estimators with a quadrature reference, crude Monte Carlo, cross-entropy, Glasserman–Li two-step sampling, conditional Monte Carlo and subset simulation at levels 10^{-3} to 10^{-6}.

Relation to the literature.

The ingredients are classical, and we use them as such. Exponential twisting at a dominating point is efficient under a geometric condition on the rare set (Sadowsky and Bucklew 1990; Bucklew 2004); a single twist fails when the rare set can be reached in several ways (Glasserman and Wang 1997); mixtures of twists are instances of the subsolution approach of Dupuis and Wang (2007); mixtures over the events of a union have a uniform variance bound when each event can be sampled exactly conditionally (Owen et al. 2019); and for factor copulas Glasserman and Li (2005) combine a factor shift with conditional twisting (see also Bassamboo et al. 2008). Our model is not covered by these results as stated: the rare set is a union of *non-convex* sets (positive parts of quadratic forms of a Gaussian vector, after a gamma quantile transform), its components overlap through shared factors, and the conditional sampling that the bound of Owen et al. needs is not available. What we add is the explicit form of the scales and modes for the pickup-and-delivery peak, the 1/4 threshold with its silent-bias mechanism, and a covering constant that turns the efficiency of a stage mixture into a computable condition. Surveys of the efficiency notions are Asmussen and Glynn (2007) and Bucklew (2004).

2 Model and spectral scales

2.1 Model

Let $u = (W_d, W_p, E_{d,1}, \ldots, E_{d,L}, E_{p,1}, \ldots, E_{p,L})$ be a standard normal vector in $\mathbb{R}^n$, $n = 2L + 2$, with density φ. For $\rho \in [0, 1)$ put

$$Z_{d,i} = \sqrt{\rho}\, W_d + \sqrt{1-\rho}\, E_{d,i}, \qquad Z_{p,i} = \sqrt{\rho}\, W_p + \sqrt{1-\rho}\, E_{p,i},$$

and $d_i = x_{d,i}(Z_{d,i})$, $p_i = x_{p,i}(Z_{p,i})$ with $x(z) = G^{-1}(\Phi(z))$ and G a gamma distribution function of shape $\kappa > 0$ and scale $\theta > 0$ (both may depend on the customer and the type). Then each d_i and p_i is gamma, deliveries are dependent through W_d, pickups through W_p, and deliveries and pickups are independent. With L_k and F as

in the introduction, the object of study is

$$p(c) = \mathbb{P}(A_c), \qquad A_c = \{F > c\} = \bigcup_{k=0}^{L} \{L_k > c\}.$$

Every customer with positive mean has a positive scale; customers with zero mean delivery or pickup contribute nothing and are dropped from the corresponding sums. To keep the notation light we assume all $2L$ scales positive.

2.2 Blocks

For a type $b \in \{d, p\}$ and a set $S \subseteq \{1, \dots, L\}$ let B_S be the $|S| \times n$ matrix with $(B_S u)_i = Z_{b,i}$, $i \in S$; its entries are nonnegative, each row has unit norm, and $B_S B_S^\top = C_S = \rho \mathbf{1}\mathbf{1}^\top + (1-\rho) I$. With $\Theta_S = \operatorname{diag}(\theta_{b,i})_{i \in S}$ define the $n \times n$ matrix $M_S = B_S^\top \Theta_S B_S$, the quadratic forms

$$h_S(u) = \tfrac{1}{2} \sum_{i \in S} \theta_{b,i} \, (Z_{b,i})_+^2 \;\le\; \tfrac{1}{2} u^\top M_S u,$$

and the scales $\lambda(S) \ge \mu(S)$, the two largest eigenvalues of $K_S = \Theta_S^{1/2} C_S \Theta_S^{1/2}$ (with $\mu(S) = 0$ if $|S| = 1$ and $\lambda(\emptyset) = \mu(\emptyset) = 0$). The nonzero eigenvalues of M_S and K_S coincide, with multiplicities.

Lemma 1 (Spectral facts) *Let $S \neq \emptyset$ and $\xi = \Theta_S^{1/2} \mathbf{1}$.*

(a) If $\rho > 0$, $\lambda(S)$ is the unique root in $((1-\rho)\max_S \theta, \infty)$ of $1 = \rho \sum_{i \in S} \theta_i / (\lambda - (1-\rho)\theta_i)$.

(b) $\rho \sum_S \theta_i + (1-\rho)\min_S \theta \;\le\; \lambda(S) \;\le\; \rho \sum_S \theta_i + (1-\rho)\max_S \theta$ and $\mu(S) \le (1-\rho)\max_S \theta$.

(c) If $S \subsetneq S'$ (same type) and $\rho > 0$, then $\lambda(S) < \lambda(S')$.

(d) Let $v \ge 0$ be a unit Perron vector of K_S. Then $e_S = B_S^\top \Theta_S^{1/2} v / \sqrt{\lambda(S)}$ is a unit eigenvector of M_S for $\lambda(S)$, $e_S \ge 0$, and $\beta_S = B_S e_S \ge 0$ satisfies $\sum_{i \in S} \theta_i \beta_{S,i}^2 = \lambda(S)$; if $\rho > 0$ then $\beta_S > 0$.

Proof $K_S = (1-\rho)\Theta_S + \rho \xi \xi^\top$. (a) For $\lambda > (1-\rho)\max\theta$, $\lambda I - K_S = (\lambda I - (1-\rho)\Theta_S)\big(I - \rho(\lambda I - (1-\rho)\Theta_S)^{-1} s s^\top\big)$, which is singular iff $1 = \rho \xi^\top (\lambda I - (1-\rho)\Theta_S)^{-1} \xi$, the stated equation. Its right side decreases from $+\infty$ to 0 on the interval, so there is exactly one root there. By Weyl's inequalities for a rank-one update (Horn and Johnson 2013, Thm. 4.3.4), K_S has at most one eigenvalue above $(1-\rho)\max\theta$; it has one, because for the coordinate vector ϵ_j of a customer with the largest scale, $\epsilon_j^\top K_S \epsilon_j = \theta_j > (1-\rho)\theta_j$. Hence that eigenvalue is $\lambda(S)$ and it is the root. (b) The lower bound is the Rayleigh quotient at $\xi/|\xi|$; the upper bound follows from $K_S \preceq (1-\rho)\max\theta \, I + \rho \xi \xi^\top$ and $|\xi|^2 = \sum \theta_i$; the bound on μ is Weyl's inequality for a rank-one update of $(1-\rho)\Theta_S$. (c) With $f_S(\lambda) = \rho \sum_{i \in S} \theta_i / (\lambda - (1-\rho)\theta_i)$, $f_{S'} > f_S$ on $((1-\rho)\max_{S'} \theta, \infty)$; at $\lambda = \lambda(S')$ this gives $f_S(\lambda(S')) < 1 = f_S(\lambda(S))$, and f_S is

decreasing, so $\lambda(S') > \lambda(S)$. (d) $M_S e_S = B_S^\top \Theta_S^{1/2} K_S v/\sqrt{\lambda} = \lambda e_S$ and $|e_S|^2 = v^\top K_S v/\lambda = 1$; nonnegativity follows from $B_S \geq 0$ and the Perron–Frobenius theorem for the nonnegative matrix K_S. Further $\beta_S = C_S \Theta_S^{1/2} v/\sqrt{\lambda}$, which is positive when $\rho > 0$ because C_S has positive entries and $v \neq 0$; and $\sum \theta_i \beta_{S,i}^2 = e_S^\top M_S e_S = \lambda$. □

2.3 Stages and end modes

Stage k has the delivery block $S_k^d = \{k+1, \dots, L\}$ and the pickup block $S_k^p = \{1, \dots, k\}$, with scales $\lambda_k^d = \lambda(S_k^d)$, $\lambda_k^p = \lambda(S_k^p)$ and second eigenvalues μ_k^d, μ_k^p. Put

$$\lambda_k = \max(\lambda_k^d, \lambda_k^p), \qquad \lambda^\star = \max_k \lambda_k = \max(\lambda_0^d, \lambda_L^p), \qquad \varrho = \frac{\min(\lambda_0^d, \lambda_L^p)}{\lambda^\star} \in (0, 1],$$

the second equality by Lemma 1(c) (or, for $\rho = 0$, by Cauchy interlacing). The two blocks of a stage act on disjoint coordinates of u, so

$$h_k(u) = h_{S_k^d}(u) + h_{S_k^p}(u)$$

is the positive-part quadratic form of stage k, with block-diagonal matrix of largest eigenvalue λ_k. For a stage we call the block attaining λ_k the dominant block H (either one in case of a tie) and the other the weak block; we write

$$\varrho_k = \frac{\min(\lambda_k^d, \lambda_k^p)}{\lambda_k}, \qquad \nu_k = \frac{\max(\mu_k^d, \mu_k^p)}{\lambda_k}, \qquad r_k = \frac{\lambda_k}{\lambda^\star}.$$

3 Two lemmas

Lemma 2 (Gamma quantile of a normal score) *Let $x(z) = G^{-1}(\Phi(z))$ for a gamma law of shape κ and scale θ. Then $x(z) = \theta z^2/2 + \theta(2\kappa - 1)\log z + O(1)$ as $z \to \infty$, and there is a constant C such that*

$$x(z) \leq \tfrac{1}{2}\theta z_+^2 + C\log(2 + z_+) + C \quad (z \in \mathbb{R}), \qquad x(z) \geq \tfrac{1}{2}\theta z^2 - C\log(2 + z) - C \quad (z \geq 0).$$

Proof Write $y = x/\theta$ and $\bar{G}(\theta y) = \Gamma(\kappa, y)/\Gamma(\kappa)$. Since $\Gamma(\kappa, y) \sim y^{\kappa-1} e^{-y}$ (Olver et al. 2010, §8.11) and $\bar{\Phi}(z) \sim \varphi(z)/z$,

$$-\log \bar{G}(\theta y) = y - (\kappa - 1)\log y + O(1), \qquad -\log \bar{\Phi}(z) = \tfrac{1}{2}z^2 + \log z + O(1).$$

At $y = y(z) = x(z)/\theta$ the two left sides are equal and $y(z) \to \infty$. The first relation gives $y/2 \leq -\log \bar{G}(\theta y) \leq 2y$ for large y, hence $z^2/8 \leq y(z) \leq 2z^2$ for large z and $\log y = 2\log z + O(1)$. Substituting, $y = z^2/2 + \log z + (\kappa - 1)\log y + O(1) = z^2/2 + (2\kappa - 1)\log z + O(1)$. So the two bounds hold for $z \geq z_0$ with some C. On $[0, z_0]$, x is continuous, hence bounded, and for $z < 0$, $0 \leq x(z) \leq x(0)$; enlarging C covers both. □

Lemma 3 (One block) *For a block S of type b let Π_S be the orthogonal projection of $\mathbb{R}^n$ onto the coordinates W_b and $E_{b,i}$, $i \in S$. With $e = e_S$, $\lambda = \lambda(S)$, $\mu = \mu(S)$ and $u \in \mathbb{R}^n$, write $a = \langle u, e \rangle$ and $u^\perp = \Pi_S u - ae$. Then*

$$h_S(u) \leq \tfrac{1}{2}\big(\lambda a_+^2 + \mu |u^\perp|^2\big).$$

Proof Since B_S only reads the coordinates in the range of Π_S, $B_S u = B_S \Pi_S u$ and $h_S(u) = h_S(\Pi_S u)$; also e lies in that range, so $u^\perp \perp e$ there. If $a \geq 0$: $h_S(u) \leq \frac{1}{2}(\Pi_S u)^\top M_S \Pi_S u = \frac{1}{2}(\lambda a^2 + u^{\perp\top} M_S u^\perp)$, the cross term vanishing because $M_S e = \lambda e \perp u^\perp$, and $u^{\perp\top} M_S u^\perp \leq \mu |u^\perp|^2$ by the Courant–Fischer theorem applied to M_S restricted to the range of Π_S. If $a < 0$: $B_S u = a B_S e + B_S u^\perp \leq B_S u^\perp$ entrywise because $B_S e \geq 0$ (Lemma 1(d)), so $(B_S u)_+ \leq (B_S u^\perp)_+$ and $h_S(u) \leq h_S(u^\perp) \leq \frac{1}{2}\mu |u^\perp|^2$. □

Since $|Z_{b,i}| \leq |u|$ (unit rows), Lemma 2 gives, for a constant C depending on the route only,

$$L_k \leq h_k(u) + C \log(2 + |u|) + C \qquad (u \in \mathbb{R}^n,\ k = 0, \dots, L). \tag{1}$$

For stage k, let e_H, e_W be the unit Perron directions of its dominant and weak blocks (orthogonal, by disjoint supports), and for $u \in \mathbb{R}^n$ let $a = \langle u, e_H \rangle$, $b = \langle u, e_W \rangle$ and $g^2 = |u|^2 - a^2 - b^2$. The two blocks read disjoint coordinates, so their vectors $u^\perp$ are orthogonal to each other and to e_H, e_W, and the sum of their squared norms is at most g^2. Applying Lemma 3 to each block,

$$2h_k(u) \leq \lambda_k \big(a_+^2 + \varrho_k b_+^2 + \nu_k g^2 \big). \tag{2}$$

4 Logarithmic asymptotics

Theorem 1 *As $c \to \infty$, for every stage k,*

$$-\frac{c}{\lambda_k} - O(\sqrt{c}) \ \leq\ \log \mathbb{P}(L_k > c) \ \leq\ -\frac{c}{\lambda_k} + O(\log c),$$

and the same holds for $\log \mathbb{P}(F > c)$ with λ_k replaced by $\lambda^\star$.

Proof *Upper bound.* By (1), on $\{|u| \leq c\}$ the event $\{L_k > c\}$ implies $h_k(u) > c' = c - C\log(2+c) - C$. Let $Z = (Z_{d,i})_{i \in S_k^d}$ and $Z' = (Z_{p,i})_{i \in S_k^p}$. Then $h_k \leq Q := \frac{1}{2} Z^\top \Theta Z + \frac{1}{2} Z'^\top \Theta' Z'$, a sum of two independent Gaussian quadratic forms with $\mathbb{E} e^{tQ} = \det(I - t K_{S_k^d})^{-1/2} \det(I - t K_{S_k^p})^{-1/2}$ for $t < 1/\lambda_k$. With $t = (1 - 1/c)/\lambda_k$ each eigenvalue factor is at least $1/c$, so $\mathbb{E} e^{tQ} \leq c^{L/2}$, and Chernoff's bound gives $\mathbb{P}(Q > c') \leq c^{L/2} e^{-tc'} = \exp(-c/\lambda_k + O(\log c))$. Finally $\mathbb{P}(|u| > c) \leq e^{-c^2/4}$ for large c.

Lower bound. Let S be the dominant block of stage k, so that $L_k \geq \sum_{i \in S} x_i(Z_i)$, and let $e = e_S$, $\beta = B_S e$, $\lambda = \lambda_k$, $J = \{i \in S : \beta_i > 0\}$. For $|u - se| \leq 1$ we have $Z_i \geq s\beta_i - 1$ (unit rows), so for $s \geq 1/\min_J \beta_i$ Lemma 2 and $x_i \geq 0$ give

$$L_k \ \geq\ \sum_{i \in J} \Big(\tfrac{1}{2}\theta_i (s\beta_i - 1)^2 - C \log(2 + s) - C \Big) \ \geq\ \tfrac{1}{2}\lambda s^2 - C' s,$$

using $\sum_J \theta_i \beta_i^2 = \lambda$. With $s = \sqrt{2c/\lambda} + K$ and $K > C'/\lambda + 1$ the right side exceeds c for large c, so $\mathbb{P}(L_k > c) \geq \mathbb{P}(|u - se| \leq 1) \geq (2\pi)^{-n/2} V_n e^{-(s+1)^2/2} = \exp(-c/\lambda - O(\sqrt{c}))$, V_n the volume of the unit ball.

Peak. $\mathbb{P}(L_k > c) \leq \mathbb{P}(F > c) \leq \sum_k \mathbb{P}(L_k > c)$, and $\max_k \lambda_k = \lambda^\star$. □

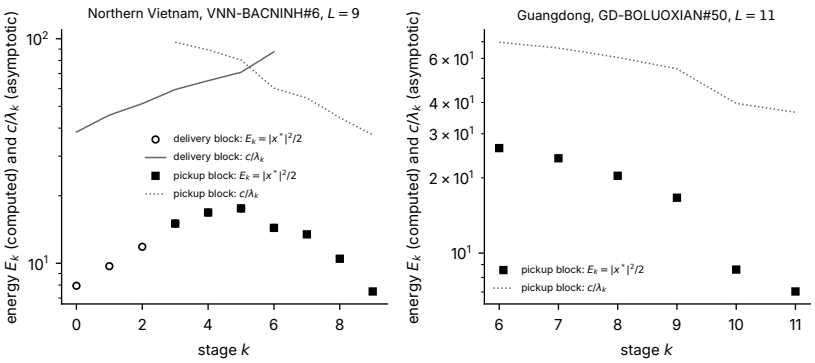


Fig. 1 Computed dominating-point energies $E_k = \min\{|u|^2/2 : L_k(u) \geq c\}$ of the stage events (open circles: delivery block; filled squares: pickup block) against the stage k, with the asymptotic value c/λ_k (lines), at level 10^{-4}. Left: a route with balanced pickups and deliveries ($\varrho = 0.93$), with two end modes of nearly equal energy. Right: a route whose pickups are 2.7 times its deliveries; no delivery block reaches c in the simulator, whose normal scores are clipped at $\Phi^{-1}(1 - 10^{-15})$.

Corollary 1 (End modes) *Let $\rho > 0$. Then λ_k^d strictly decreases and λ_k^p strictly increases in k, so $\lambda_k < \lambda^\star$ for $0 < k < L$, and for each such stage $\mathbb{P}(L_k > c)/\mathbb{P}(F > c) = \exp\{-c(1/\lambda_k - 1/\lambda^\star) + O(\sqrt{c})\}$. If $\varrho < 1$, the weaker end event has relative mass $\exp\{-c(1/(\varrho\lambda^\star) - 1/\lambda^\star) + O(\sqrt{c})\}$; at a level $\gamma = \mathbb{P}(F > c)$ this is $\gamma^{1/\varrho - 1 + o(1)}$.*

Proof Lemma 1(c) and Theorem 1; $c = \lambda^\star \log(1/\gamma)(1 + o(1))$. □

Remark 1 (Pre-asymptotic regime) The error terms are not small at practical levels. For the coefficient of variation 0.3 used below the gamma shape is $\kappa \approx 11.1$, and the second term $\theta(2\kappa - 1)\log z$ of Lemma 2 is larger than the leading term $\theta z^2/2$ for $1.1 \leq z \leq 9.9$, a range that contains every normal score relevant at levels 10^{-3} to 10^{-6}. Accordingly the exact minimum of $|u|^2/2$ over $\{L_k \geq c\}$, computed numerically, is 5 to 10 times smaller than c/λ_k on our routes (Figure 1). The theorem identifies the modes and their ordering; the estimators of Section 7 place their shifts at computed minimisers, and the masses of modes are compared through the computed energies.

5 A single mean shift

Importance sampling with proposal q estimates $p = \mathbb{P}(A)$ by $\hat{p}_N = N^{-1}\sum_{j \leq N} \mathbf{1}_A(U_j)\,\varphi(U_j)/q(U_j)$, $U_j \sim q$ i.i.d.; its second moment per draw is $m_2 = \int_A \varphi^2/q$. The estimator is *logarithmically efficient* if $\log m_2/(2\log p) \to 1$ (Asmussen and Glynn 2007). For the mean shift $q = \varphi(\cdot - \tau)$ the likelihood ratio is $w(u) = e^{-\langle u,\tau\rangle + |\tau|^2/2}$. All our proposals are mixtures $q = \sum_m a_m \varphi(\cdot - \tau_m)$, $a_m > 0$, $\sum a_m = 1$,

and we bound m_2 through

$$J_m(u) = \tfrac{1}{2}|u+\tau_m|^2 - |\tau_m|^2, \qquad \Upsilon(u) = \max_m \big(J_m(u) + \log a_m\big).$$

Lemma 4 (Second moment of a mixture) *$m_2 \le (2\pi)^{-n/2}\int_A e^{-\Upsilon(u)}du$. If $\Upsilon \ge \gamma_0 > 0$ on A, then for every component m, $m_2 \le a_m^{-1} C_n\,(1+\gamma_0+|\tau_m|^2)^{n/2}e^{-\gamma_0}$ with C_n depending on n only.*

Proof $\varphi^2/q \le \min_m \varphi^2/(a_m\varphi(\cdot-\tau_m))$ and $\varphi(u)^2/\varphi(u-\tau) = (2\pi)^{-n/2}e^{-J(u)}$, which gives the first bound. Fix m. On A, $\Upsilon \ge \max(\gamma_0, J_m + \log a_m) \ge \max(\gamma_0, J_m) + \log a_m$, so $m_2 \le a_m^{-1}(2\pi)^{-n/2}e^{|\tau_m|^2}\int e^{-\max(\gamma_0+|\tau_m|^2, |v|^2/2)}dv$ after the substitution $v = u+\tau_m$. With $R^2 = 2(\gamma_0+|\tau_m|^2)$, the integral is at most $V_n R^n e^{-R^2/2} + (2\pi)^{n/2}\mathbb{P}(\chi^2_n > R^2)$, and both terms are $O((1+R^2)^{n/2}e^{-R^2/2})$. □

Write $\lambda_{\mathrm{o}} = \varrho\lambda^\star$ for the weaker end scale and let $e^\star$ be the Perron direction of the dominant end block, $\mu^\star$ its second eigenvalue, and $\tau^\star = \sqrt{2c/\lambda^\star}\,e^\star$ the asymptotic dominating point.

Theorem 2*(a) Let τ be any vector supported on the inputs of the dominant end's type (the delivery inputs $W_d, E_{d,\cdot}$ when the dominant end is the delivery block; for instance $\tau^\star$). Then $\log m_2 \ge |\tau|^2 - c/\lambda_{\mathrm{o}} - O(\sqrt{c})$. For $\tau = \tau^\star$,*

$$\log m_2 - 2\log p \;\ge\; \frac{c}{\lambda^\star}\Big(4 - \frac{1}{\varrho}\Big) - O(\sqrt{c}),$$

so the shift $\tau^\star$ is not logarithmically efficient if $\varrho > 1/4$.

(b) For the defensive version $q = a_0\varphi + (1-a_0)\varphi(\cdot-\tau^\star)$ with a fixed $a_0 \in (0,1)$, $\log m_2 \ge -c/\lambda_{\mathrm{o}} - O(\sqrt{c})$, so it is not logarithmically efficient if $\varrho > 1/2$.

(c) If $\varrho \le 1/4$ and $\mu^\star \le \lambda^\star/4$, the shift $\tau^\star$ is logarithmically efficient; indeed $\log m_2 \le -2c/\lambda^\star + O(\log c)$.

Proof Say the dominant end is the delivery block at stage 0; the other case is symmetric. Let $\mathcal{B} = \{P_L > c\} \subseteq A$, an event of the pickup inputs only, with $\log\mathbb{P}(\mathcal{B}) = -c/\lambda_{\mathrm{o}} + O(\sqrt{c})$ by Theorem 1.

(a) w depends on the delivery inputs only, so by independence

$$m_2 \ge \mathbb{E}[w\mathbf{1}_{\mathcal{B}}] = \mathbb{E}[w]\,\mathbb{P}(\mathcal{B}) = e^{|\tau|^2}\mathbb{P}(\mathcal{B}).$$

With $|\tau^\star|^2 = 2c/\lambda^\star$ and $\log p = -c/\lambda^\star + O(\sqrt{c})$ the display follows, and then $\limsup \log m_2/(2\log p) \le 1 - (4-1/\varrho)/2 < 1$ when $\varrho > 1/4$.

(b) Here $w = 1/(a_0 + (1-a_0)e^{\langle u,\tau^\star\rangle - |\tau^\star|^2/2}) \ge 1$ on $\{\langle u,\tau^\star\rangle \le |\tau^\star|^2/2\}$, an event of the delivery inputs of probability at least $1/2$; by independence $m_2 \ge \frac{1}{2}\mathbb{P}(\mathcal{B})$. Efficiency would need $\log m_2 \le -2c/\lambda^\star + o(c)$, which fails when $1/\lambda_{\mathrm{o}} < 2/\lambda^\star$.

(c) Since $F \le D_0 + P_L$, Lemma 2 gives $A \subseteq \{h_{S_0^d}(u) + h_{S_L^p}(u) \ge c - C\log(2+|u|) - C\}$. Apply Lemma 3 to the dominant block with $a = \langle u, e^\star\rangle$, and bound the other block, of matrix M_o and projection Π_o, by $h \le \frac{1}{2}u^\top M_o u \le \frac{1}{2}\lambda_o|\Pi_o u|^2$; the two blocks read disjoint coordinates, so $|u^\perp|^2 + |\Pi_o u|^2 \le |u|^2 - a^2$. With $g^2 = |u|^2 - a^2$ and $\nu' = \max(\mu^\star, \lambda_o)/\lambda^\star \le 1/4$,

$$2(h_{S_0^d} + h_{S_L^p})(u) \le \lambda^\star(a_+^2 + \nu' g^2).$$

The single component has $J(u) = \frac{1}{2}|u+\tau^\star|^2 - |\tau^\star|^2$. Let $R = 4|\tau^\star|$. If $|u| > R$ then $J(u) \ge \frac{1}{2}(3|\tau^\star|)^2 - |\tau^\star|^2 > 2c/\lambda^\star$. If $|u| \le R$, then $u \in A$ implies $a_+^2 + \nu' g^2 \ge 2c'/\lambda^\star$ with $c' = c - O(\log c)$. Scale $y = u\sqrt{\lambda^\star/c'}$, so that $a_{y,+}^2 + \nu' g_y^2 \ge 2$, $|y| = O(1)$ and $J(u) = \frac{c'}{2\lambda^\star}\big(|y+\sigma\sqrt{2}e^\star|^2 - 4\sigma^2\big)$ with $\sigma^2 = c/c' = 1 + O(\log c/c)$. We claim $|y+\sqrt{2}e^\star|^2 - 4 \ge 4$ under the constraint. If $a_y \ge 0$, then $g_y^2 \ge 4(2 - a_y^2)_+$ and $|y+\sqrt{2}e^\star|^2 - 4 = a_y^2 + 2\sqrt{2}a_y - 2 + g_y^2 \ge 6 - 3a_y^2 + 2\sqrt{2}a_y \ge 4$ for $a_y \le \sqrt{2}$ (a concave function equal to 6 and 4 at the endpoints), and $\ge (2\sqrt{2})^2 - 4 = 4$ for $a_y > \sqrt{2}$. If $a_y < 0$, then $g_y^2 \ge 8$ and the left side is at least $g_y^2 - 4 \ge 4$. Replacing $\sqrt{2}$ by $\sigma\sqrt{2}$ changes the left side by $O(\sigma - 1)$ on $|y| = O(1)$, so $J \ge 2c/\lambda^\star - O(\log c)$ on A, and Lemma 4 gives the bound. Since $m_2 \ge p^2$, efficiency follows from Theorem 1. □

The threshold 1/4 in (a) and (c) is sharp in ϱ; the extra condition on $\mu^\star$ in (c) is a spectral gap of the dominant block, which by Lemma 1(b) holds when $\rho\sum\theta_i + (1-\rho)\min\theta \ge 4(1-\rho)\max\theta$. The proof of (a) also shows where the second moment comes from: draws with typical delivery inputs and pickup inputs in $\mathcal{B}$, which carry weight $e^{-\langle u,\tau\rangle + |\tau|^2/2}$ but are proposed no more often than under crude Monte Carlo. That has a consequence for what a user sees.

Proposition 1 (Silent underestimation) *Let q be any proposal under which the pickup inputs keep their standard normal law, independent of the delivery inputs (for instance a shift supported on the delivery inputs, with or without a defensive component, or the two-step scheme of Glasserman and Li (2005) with its factor and inner shifts taken from the delivery end mode, whose dominating point has no pickup coordinates), and let $\mathcal{B} = \{P_L > c\}$, $p_{\mathcal{B}} = \mathbb{P}(\mathcal{B})$. Then $q(\mathcal{B}) = p_{\mathcal{B}}$, and for the estimator $\hat{p}_N$,*

$$\mathbb{P}_q(\text{no draw in } \mathcal{B}) = (1-p_{\mathcal{B}})^N \ge 1 - Np_{\mathcal{B}}, \qquad \mathbb{E}_q\big[\hat{p}_N \,\big|\, \text{no draw in } \mathcal{B}\big] = \frac{p - p_{\mathcal{B}}}{1 - p_{\mathcal{B}}}.$$

So on an event of probability at least $1 - Np_{\mathcal{B}}$ the estimator has conditional mean $p(1-s)(1+O(p))$, where $s = p_{\mathcal{B}}/p$ is the share of the weaker end event, and the sample variance it reports is computed from draws that all avoid $\mathcal{B}$, so it estimates the variance under $q(\cdot \mid \mathcal{B}^c)$, not m_2. The symmetric statement holds with deliveries and pickups exchanged.

Proof $q(\mathcal{B}) = p_{\mathcal{B}}$ because $\mathcal{B}$ is an event of the pickup inputs. Given that no draw falls in $\mathcal{B}$ the draws are i.i.d. from $q(\cdot \mid \mathcal{B}^c)$, and $\mathbb{E}_q[\mathbf{1}_A w \mid \mathcal{B}^c] = \int_{A\setminus\mathcal{B}} \varphi/(1-p_{\mathcal{B}}) = (p - p_{\mathcal{B}})/(1-p_{\mathcal{B}})$, because $\mathcal{B} \subseteq A$. Finally $(p-p_{\mathcal{B}})/(1-p_{\mathcal{B}}) = p(1-s)(1+O(p))$. □

The proposition explains three facts reported in Section 7: the single shift underestimates by about the share s of the weaker end mode; its reported standard error does not reveal this, because the variance it estimates is that of a sample that avoids $\mathcal{B}$; and Glasserman–Li two-step sampling, which shifts the dominant factor only, has the same bias. Unbiasedness is restored only through rare draws in $\mathcal{B}$, each carrying

a weight of order $e^{-\langle u,\tau\rangle+|\tau|^2/2}$. By Corollary 1, $s=\exp\{-c(1/\lambda_{\mathrm{o}}-1/\lambda^\star)+O(\sqrt{c})\}$, so the bias is negligible for small ϱ and of order one for ϱ near 1, while the threshold of Theorem 2(a) is 1/4: variance inflation and visible bias are different regimes.

6 Stage mixtures

A mixture places components at the dominating points of several stage blocks. For a stage k and a block b of it, the asymptotic dominating point of $\{h_{S_k^b}\ge c\}$ is $\tau_{k,b}=\sqrt{2c/\lambda_k^b}\,e_{S_k^b}$, and we consider

$$q=a_0\varphi+\sum_{m\in\mathcal{M}}a_m\varphi(\cdot-\tau_m),\qquad a_0\in(0,1)\text{ fixed},\quad a_m=e^{-c\delta_m+o(c)},\ \delta_m\ge 0,$$

for a set $\mathcal{M}$ of (stage, block) pairs. Constant weights have $\delta_m=0$; weights proportional to the asymptotic mode masses, $a_m\propto e^{-c/\lambda_m}$, have $\delta_m=1/\lambda_m-1/\lambda^\star$.

Definition 1 (Covering constant) For $\varrho\in[0,1]$, $\nu\in[0,1]$ and penalties $\pi_H,\pi_W\in[0,\infty]$ let

$$\begin{aligned}F_0(a,b,g)&=a^2+b^2+g^2,\\ F_H(a,b,g)&=(a+\sqrt{2})^2+b^2+g^2-4-\pi_H,\\ F_W(a,b,g)&=a^2+\bigl(b+\sqrt{2/\varrho}\bigr)^2+g^2-4/\varrho-\pi_W,\end{aligned}$$

and $\Psi(\varrho,\nu;\pi_H,\pi_W)=\inf\{\max(F_0,F_H,F_W):\ a,b\in\mathbb{R},\ g\ge 0,\ a_+^2+\varrho b_+^2+\nu g^2\ge 2\}$, with F_W omitted when $\varrho=0$ or $\pi_W=\infty$ and F_H omitted when $\pi_H=\infty$. We write $\psi(\varrho,\nu)=\Psi(\varrho,\nu;0,0)$ (both blocks, constant weights) and $\psi_H(\varrho,\nu)=\Psi(\varrho,\nu;0,\infty)$ (dominant block only).

Every function in the maximum increases in g, so for $\nu>0$ the infimum over g is attained at $g^2=(2-a_+^2-\varrho b_+^2)_+/\nu$ (for $\nu=0$ the constraint forces $a_+^2+\varrho b_+^2\ge 2$ and $g=0$), and Ψ is the minimum of an explicit function of two variables; it is evaluated numerically in seconds.

Lemma 5 *Let $\varrho,\nu\in[0,1]$.*

(a) $2\le\Psi\le 4$ whenever $\pi_H=0$; with no components ($\pi_H=\pi_W=\infty$) $\Psi\ge 2$.
(b) If $\varrho\le 1/3$ and $\nu\le 1/3$, then $\psi_H(\varrho,\nu)=\psi(\varrho,\nu)=4$.
(c) If $\nu\le 1/2$, then $\psi(\varrho,\nu)\ge 2\sqrt{2}$, with equality at $\varrho=1$.
(d) For a stage with one block only ($\varrho=0$) and $\nu\in(0,1]$, $\psi_H(0,\nu)=\min\bigl(4,\frac{1}{2}+\frac{3}{2\nu}\bigr)$; it equals 4 iff $\nu\le 3/7$.

Proof (a) $F_0\ge a_+^2+\varrho b_+^2+\nu g^2\ge 2$. At $(a,b,g)=(\sqrt{2},0,0)$, $F_0=2$, $F_H=4$ and $F_W=2-2/\varrho\le 0$.

In (b) and (c) write $X=a_+^2$, $Y=\varrho b_+^2$, $W=\nu g^2$, so that $X+Y+W\ge 2$, and note $b^2\ge Y/\varrho$, $g^2=W/\nu$.

(b) If $a > 0$: $F_H \geq X + 2\sqrt{2X} - 2 + 3(Y + W) \geq X + 2\sqrt{2X} - 2 + 3(2 - X)_+$, which is at least 4: for $X \leq 2$ it is the concave function $4 - 2X + 2\sqrt{2X}$, equal to 4 at both ends, and for $X > 2$ it exceeds $2 + 4 - 2$. If $a \leq 0$: $Y + W \geq 2$ and $F_0 \geq b^2 + g^2 \geq 3(Y + W) \geq 6$. With (a), $\psi_H = 4$; adding F_W cannot lower the maximum, and $\psi \leq 4$ by (a).

(c) Here $g^2 \geq 2W$, and let $y_0 = (2 - \sqrt{2})^2$, the root of $Y + 2\sqrt{2Y} = 2$; the monotonicity in X, Y, W allows us to assume $X + Y + W = 2$. *Case* $a, b > 0$. $F_0 \geq X + Y + 2W$, $F_H \geq X + 2\sqrt{2X} - 2 + Y + 2W$ and $F_W = X + g^2 + (Y + 2\sqrt{2Y} - 2)/\varrho$. If $Y \geq y_0$, the last bracket is nonnegative and $F_W \geq X + 2\sqrt{2Y} - 2 + Y + 2W$; with $S = X + Y$ and $\max(\sqrt{X}, \sqrt{Y}) \geq \sqrt{S/2}$, $\max(F_0, F_H, F_W) \geq \max(4 - S, 2 - S + 2\sqrt{S})$, which is at least 3 for $S \leq 1$ and at least $2\sqrt{2}$ on $[1, 2]$ (concavity; the value at $S = 2$ is $2\sqrt{2}$). If $Y < y_0$: either $W \geq 2\sqrt{2} - 2$, and $F_0 \geq 2 + W \geq 2\sqrt{2}$, or $X \geq 2 - y_0 - W$ and $F_H \geq W + 2\sqrt{2X} \geq W + 2\sqrt{2(2 - y_0 - W)}$, a concave function of $W \in [0, 2\sqrt{2} - 2]$ whose end values are 3.64 and 3.40. *Case* $a \leq 0 < b$. Then $Y + W = 2$ and, dropping a^2, $F_W \geq (Y + 2\sqrt{2Y} - 2)/\varrho + 2W$. If $Y \geq y_0$, $F_W \geq 2 - Y + 2\sqrt{2Y} \geq 3.31$ on $[y_0, 2]$; if $Y < y_0$, $F_0 \geq Y + 2W = 4 - Y > 3.6$. *Case* $b \leq 0 < a$ is the same with F_H. *Case* $a, b \leq 0$. $F_0 \geq g^2 \geq 2/\nu \geq 4$. Equality at $\varrho = 1$: at $a = b = 1$, $g = 0$ we get $F_0 = 2$ and $F_H = F_W = 2\sqrt{2}$.

(d) With $\varrho = 0$ the constraint is $a_+^2 + \nu g^2 \geq 2$ (the coordinate b is part of g). If $a \leq 0$, $F_0 \geq g^2 \geq 2/\nu \geq \frac{1}{2} + \frac{3}{2\nu}$. If $a > 0$, put $X = a^2$; both functions increase in g, so take $g^2 = (2 - X)_+/\nu$. Then $F_H \geq F_0$ iff $X \geq 1/2$. On $[0, 1/2]$ the maximum is $F_0 = X + (2 - X)/\nu$, decreasing in X, with value $\frac{1}{2} + \frac{3}{2\nu}$ at $X = 1/2$; on $[1/2, 2]$ it is $F_H = X(1 - 1/\nu) + 2\sqrt{2X} - 2 + 2/\nu$, concave, with end values $\frac{1}{2} + \frac{3}{2\nu}$ and 4; for $X > 2$, $F_H > 4$. The infimum is the smaller end value, attained. □

Theorem 3 *For each stage k let $\pi_H = 2\lambda_k \delta_m$ if the dominant block $m = (k, H)$ of the stage belongs to $\mathcal{M}$ and $\pi_H = \infty$ otherwise, π_W likewise for the weak block, and $\Psi_k = \Psi(\varrho_k, \nu_k; \pi_H, \pi_W)$. If*

$$\Psi_k \;\geq\; 4\, r_k = 4\lambda_k/\lambda^\star \qquad \text{for every stage } k = 0, \ldots, L, \tag{3}$$

then $\log m_2 \leq -2c/\lambda^\star + o(c)$, and the mixture estimator is logarithmically efficient. With constant weights the error term is $O(\log c)$.

Proof By Lemma 4 applied with the defensive component $m = 0$ (whose weight a_0 is fixed) it suffices to show $\Upsilon(u) \geq 2c/\lambda^\star - o(c)$ on A. Since $\Upsilon \geq J_0 + \log a_0 = |u|^2/2 + \log a_0$, this holds for $|u| > R := 3\sqrt{c/\lambda^\star}$, and we take $u \in A$ with $|u| \leq R$. Then $u \in \{L_k > c\}$ for some k, and by (1) and (2), $a_+^2 + \varrho_k b_+^2 + \nu_k g^2 \geq 2c'/\lambda_k$ with $c' = c - O(\log c)$, in the coordinates of stage k. Put $y = u\sqrt{\lambda_k/c'}$ and $\sigma^2 = c/c'$. In these coordinates the components of the stage are $\tau_{k,H} = \sqrt{c'/\lambda_k}\,\sigma\sqrt{2}\, e_H$ and $\tau_{k,W} = \sqrt{c'/\lambda_k}\,\sigma\sqrt{2/\varrho_k}\, e_W$, and

$$J_0(u) = \frac{c'}{2\lambda_k} F_0(y), \qquad J_{k,H}(u) + \log a_{k,H} = \frac{c'}{2\lambda_k} F_H^\sigma(y) + o(c),$$

where F_H^σ is F_H with $\sqrt{2}$ replaced by $\sigma\sqrt{2}$, 4 by $4\sigma^2$ and $\pi_H = 2\lambda_k\delta_{k,H}$ by $2\lambda_k\delta_{k,H}c/c'$; likewise for W. Since $|y| = O(1)$ and $\sigma \to 1$, $F^\sigma \geq F - o(1)$ uniformly, so $\Upsilon(u) \geq \frac{c'}{2\lambda_k}(\Psi_k - o(1)) \geq \frac{2c}{\lambda^\star} - o(c)$ by (3). With $\delta_m = 0$ the $o(\cdot)$ terms are $O(\log c)$, as in the proof of Theorem 2(c). □

Corollary 2 *With constant weights, (3) holds at stage k if (i) $r_k \le 1/2$, with no component for the stage; or (ii) $\varrho_k \le 1/3$, $\nu_k \le 1/3$ and the dominant block of the stage is in $\mathcal{M}$; or (iii) $r_k \le 1/\sqrt{2}$, $\nu_k \le 1/2$ and both blocks of the stage are in $\mathcal{M}$.*

Proof Lemma 5(a), (b) and (c). □

Remark 2 (1) The end stages need components. Stage 0 has an empty pickup block ($\varrho_0 = 0$, $r_0 = \lambda_0^d/\lambda^\star$), so by Lemma 5(d) the delivery end component covers it iff $\frac{1}{2} + \frac{3}{2\nu_0} \ge 4r_0$; at the dominant end ($r_0 = 1$) this is the gap condition $\nu_0 \le 3/7$. The same holds at stage L. For mass weights, at an end stage with no weak block and $\nu_k \le 1/3$, the proof of Lemma 5(b) gives $F_H - \pi_H \ge 4 - 2(1 - r_k) = 2 + 2r_k \ge 4r_k$ where F_H covers, and $F_0 \ge 6$ elsewhere; so the penalties do not hurt at the ends. (2) Interior stages close to an end have ϱ_k small and are covered by (ii); the crossing stage, where the two blocks are comparable, has r_k near $1/2$ on routes with many stops, because the scales are close to $\rho \sum \theta$ and the stage splits the route (Lemma 1(b)). For ties with $r_k > 1/\sqrt{2}$ the covering inequality fails with two components and constant weights ($\psi = 2\sqrt{2} < 4r_k$ by Lemma 5(c)), so Theorem 3 gives no guarantee; a continuum of components along the arc between the two block directions is the natural remedy, in the spirit of Dupuis and Wang (2007). We do not pursue this, because it does not occur on our routes at $\rho \ge 0.6$ (Table 1). (3) The theorem is stated for the asymptotic dominating points. Its proof uses the shifts only through $\tau_m/\sqrt{c}$, so it remains true for any shifts with $\tau_m(c) = \tau_m + o(\sqrt{c})$. We do not prove that the computed minimisers used in practice satisfy this.

Corollary 3 (Several routes) *Let routes $j = 1, \dots, J$ share the factors W_d, W_p and have their own idiosyncratic inputs, with capacities $c_j = \alpha_j c$ for fixed $\alpha_j > 0$. Then $\log \mathbb{P}(\bigcup_j \{F_j > c_j\}) = -c \min_j \alpha_j/\lambda_j^\star + O(\sqrt{c})$, and the mixture over the stage components of all routes is logarithmically efficient if (3) holds for every route and stage with $\lambda^\star$ replaced by $\max_j \lambda_j^\star/\alpha_j$ and λ_k by $\lambda_{j,k}/\alpha_j$.*

Proof Each stage event involves one route, and the input coordinates of its two blocks are still disjoint from each other; the proofs of Theorems 1 and 3 apply to the finite union over (route, stage) pairs, after rescaling the capacities. □

7 Numerical illustration

7.1 Setting

We use the routes of plans for two regions, built by an adaptive large-neighbourhood search on road-network instances: 26 depot areas in Northern Vietnam (375 routes, 2 612 customers, capacity 15 t) and 122 in Guangdong (5 867 routes, 37 487 customers, 30 t). The mean demands come from regional parcel statistics; in Vietnam the pickup rate equals the delivery rate (no regional figure is published), in Guangdong pickups are on average 2.7 times the deliveries. Routes have 6–7 stops in median and up to 40. The demand model is that of Section 2 with coefficient of variation 0.3 and $\rho = 0.6$, and we vary ρ and the coefficient of variation in Section 7.4. Normal scores are clipped

at $\Phi^{-1}(1-10^{-15})$ in the simulator; this changes each input's law on a set of probability 10^{-15} and every probability below by less than 10^{-13}. The two regions differ in ϱ: median 0.93 in Vietnam (balanced routes) and 0.36 in Guangdong.

Estimators. SMIS is the stage mixture of Section 6 with three practical choices: the shifts are computed minimisers of $|u|^2/2$ over $\{L_k \geq c\}$ (SLSQP, started on the block's Perron ray, with the exact quantile transform); a mode is kept if its computed energy E_m is within $\log(1/\eta)$ of the smallest, $\eta = 10^{-3}$; and the weights are $a_m \propto e^{-(E_m - E_{\min})}$, the finite-level analogue of mass weights, with a defensive weight $a_0 = 0.05$. There is no pilot run. We compare it with the single shift at the dominant end tuned by the multilevel cross-entropy (CE) method (de Boer et al. 2005), a CE mixture with one CE shift per stage, Glasserman–Li two-step sampling (Glasserman and Li 2005) with the factor shift at the dominant point and an inner exponential shift, conditional Monte Carlo over the two factors with an inner dynamic program, and subset simulation (Au and Beck 2001). The reference is a quadrature program: given the two factors the demands are independent, so $\mathbb{P}(F > c \mid W_d, W_p)$ is a one-dimensional dynamic program on a discretised load (256 and 512 levels with Richardson extrapolation), integrated by a product trapezoid rule on $[-6, 9]^2$; its error estimate is below 1% of p at the grids used. Capacities are set so that $p \approx \gamma \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$.

7.2 The conditions on real routes

Table 1 evaluates the conditions of Theorems 2 and 3 on all 5 920 routes with at least two stops. Every route has $\varrho > 1/4$, so the single shift is never efficient: the threshold of Theorem 2 is not informative about which routes fail in practice; the share s of Proposition 1 is. With constant weights the covering condition (3) holds for every route at $\rho = 0.6$ and 0.8, with a smallest margin $\Psi_k - 4r_k$ of 0.23 at $\rho = 0.6$ among the stages that need the exact constant; the closed-form conditions of Corollary 2 miss three quarters of the Vietnamese routes, whose crossing stages have $r_k \in (0.71, 0.90)$ and $\varrho_k \in (0.33, 0.8)$. At $\rho = 0.3$ the blocks have smaller spectral gaps and the condition holds for 86% of the routes. With mass weights, the finite-level analogue of which SMIS uses, it holds for every Guangdong route at $\rho \geq 0.6$ but for only 28% of the Vietnamese routes at $\rho = 0.6$: on balanced routes the crossing stages carry a weak-block penalty that the covering constant cannot absorb. Section 7.4 shows that SMIS is accurate on these routes too, so the condition is sufficient, not necessary; for a guarantee on balanced routes, constant weights on the crossing-stage components are the safe choice.

7.3 The single shift and the missed end mode

On 120 routes (80 from Guangdong, 40 from Vietnam, sampled with a fixed seed and stratified by number of stops; 4 to 15 stops) and four levels, every estimator uses 10^6 draws. Table 2 reports the relative error against the quadrature reference where it is available (routes with at most eight stops at 10^{-3} and 10^{-4}) and against the CE mixture otherwise. On routes with $\varrho \leq 0.5$ the single shift is as accurate as the mixtures; on the balanced routes ($\varrho > 0.75$) its median error is 3–6%, with 90% quantiles up to 40%, and it underestimates in 93–95% of the rows at $\gamma \leq 10^{-4}$. Its

Table 1 Conditions of the theorems on all routes with at least two stops: median end-mode ratio ϱ, share of routes with $\varrho > 1/4$ (single shift not efficient, Theorem 2), and share satisfying the covering condition (3) with components for both blocks of every stage with $r_k > 1/2$: by the closed-form cases of Corollary 2, with the exact constant ψ and constant weights, and with mass weights $\delta_m = 1/\lambda_m - 1/\lambda^\star$. The conditions do not depend on the coefficient of variation.

ρ	region	routes	ϱ	$\varrho > 1/4$	Cor. 2	ψ, constant	Ψ, mass
0.3	Vietnam	347	0.92	100.0%	0.0%	91.6%	0.3%
0.3	Guangdong	5573	0.36	100.0%	3.3%	85.3%	84.0%
0.6	Vietnam	347	0.93	100.0%	23.9%	100.0%	27.7%
0.6	Guangdong	5573	0.36	100.0%	99.9%	100.0%	100.0%
0.8	Vietnam	347	0.93	100.0%	52.4%	100.0%	63.7%
0.8	Guangdong	5573	0.37	100.0%	100.0%	100.0%	100.0%

Table 2 120 routes, 10^6 draws: absolute relative error (median / 90% quantile) and sample relative variance (median) by band of ϱ. The reference is the quadrature program for 178 of the 480 rows and the CE mixture for the others, so the CE mixture has no entry at $\gamma \le 10^{-5}$. One route in $(0.5, 0.75]$ is omitted.

γ	ϱ	n	SMIS	CE mixture	single shift	RV SMIS	RV single
10^{-3}	≤ 0.5	79	0.14 / 0.38%	0.09 / 0.33%	0.18 / 0.45%	3.77	3.51
	> 0.75	40	0.22 / 0.44%	0.05 / 0.32%	2.86 / 7.57%	4.05	512
10^{-4}	≤ 0.5	79	0.17 / 0.46%	0.13 / 0.40%	0.18 / 0.48%	4.51	4.23
	> 0.75	40	0.20 / 0.42%	0.07 / 0.36%	5.10 / 21.7%	4.71	87
10^{-5}	≤ 0.5	79	0.20 / 0.48%	ref.	0.16 / 0.51%	5.18	4.88
	> 0.75	40	0.26 / 0.58%	ref.	6.00 / 26.7%	5.31	5.22
10^{-6}	≤ 0.5	79	0.26 / 0.56%	ref.	0.23 / 0.56%	5.79	5.46
	> 0.75	40	0.22 / 0.52%	ref.	6.05 / 39.9%	5.88	5.50

sample relative variance reveals the problem at 10^{-3} and 10^{-4} (512 and 87, against 4–5 for SMIS) but not at 10^{-5} and 10^{-6} (5.2 and 5.5), as Proposition 1 predicts once Np_B is small. The relative variance of SMIS grows from 3.8 to 5.9 over three decades of γ, consistent with logarithmic efficiency.

The share predicts the error. Proposition 1 says that, on the typical event, the single shift and the Glasserman–Li estimator have mean $p(1 - s)$, with $s = \mathbb{P}(\text{weaker end event})/p$. We estimated both end probabilities for the 30 routes of Table 4 at 10^{-4} and 10^{-6} on the single blocks (the route with its pickups, respectively deliveries, set to zero), where there is one mode and SMIS has a relative standard error of about 0.2% with 10^6 draws (Table 3). In the 47 route–level rows where an estimator expects fewer than one draw in the weaker mode ($Np_{\mathcal{B}} < 1$), its underestimate follows s: the Spearman correlation is 0.92 and the median ratio of underestimate to s is 1.00. The single shift at 10^{-4} expects about 14 such draws in median; it then reaches the weaker mode, and the deficit shows partly as error and partly as variance (relative variance 212 in Table 4). On the routes with $\varrho \le 0.75$, $s < 10^{-15}$ and both estimators are unbiased within their sampling error. A cheaper proxy, available without quadrature, is $\hat{m} = \exp\{-(E_o - E_{\min})\}$ with the computed energies of the

Table 3 Silent underestimation on the 15 balanced routes ($\varrho > 0.75$) of Table 4. $p_{\mathcal{B}}$ is the probability of the weaker end event, estimated on the single block, and $s = p_{\mathcal{B}}/p$ its share, with p the quadrature reference; $Np_{\mathcal{B}}$ is the expected number of draws that reach that mode. Underestimate: $-(\hat{p} - p)/p$. Ratio: underestimate/s on the rows with $s > 0.01$. All medians over routes; last column: routes with $Np_{\mathcal{B}} < 1$.

estimator	γ	N	$Np_{\mathcal{B}}$	s	underestimate	ratio	$Np_{\mathcal{B}} < 1$
single CE shift	10^{-4}	10^6	14	0.139	6.2%	0.90	2
single CE shift	10^{-6}	10^6	0.085	0.086	9.7%	1.02	15
Glasserman–Li	10^{-4}	2×10^4	0.28	0.139	15.5%	1.05	15
Glasserman–Li	10^{-6}	2×10^4	0.0017	0.086	7.4%	0.97	15

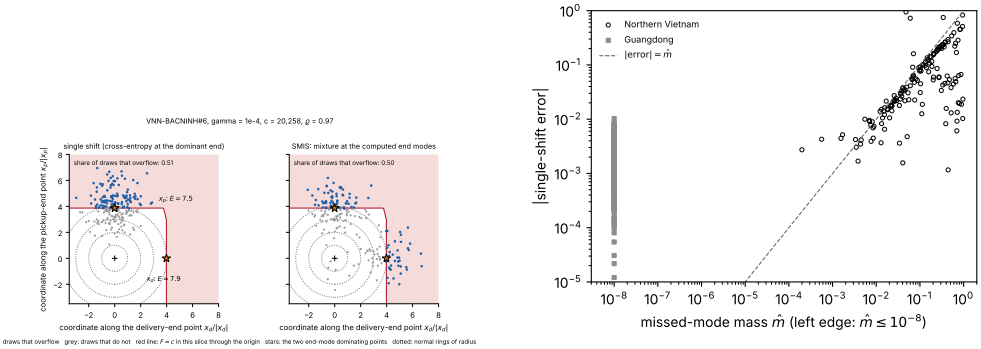


Fig. 2 Left: the two end modes of a balanced route at 10^{-4} in the plane of the two computed dominating points (shaded: the overflow set in this slice; stars: the two dominating points); a single shift at one star reaches the other mode only as often as crude sampling does. Right: absolute error of the single CE shift against the computed weaker-mode mass $\hat{m}$, all 480 route–level rows (circles: Vietnam; squares: Guangdong, all with $\hat{m} \le 10^{-8}$, drawn at the left edge), with the line $|\text{error}| = \hat{m}$.

two end modes; over all 480 route–level rows the absolute error of the single shift is at the noise level when $\hat{m} < 10^{-6}$ (all Guangdong) and about $0.8\,\hat{m}$ when $\hat{m} \ge 10^{-2}$ (Figure 2). The retention rule of SMIS keeps exactly the modes with $\hat{m} \ge \eta$, so by the same proposition the modes it drops carry a share of order η at most.

7.4 Rivals, longer routes, coverage

Table 4 compares all methods on 30 of the routes (15 with $\varrho > 0.75$) against the quadrature reference. SMIS agrees with the reference within 0.58% on every route at both levels. Both factor-based rivals inherit the missed mode: on the balanced routes the Glasserman–Li estimator is off by 15% (10^{-4}) and 6.6% (10^{-6}) in median even with 10^6 draws, where its sampling error is about 0.5%, and the relative variance of conditional Monte Carlo jumps from about 1 to 57–153. Subset simulation needs no mode information but has a relative variance of 700–4 400 per draw. The quadrature program itself needs about 38 s per route and level at the grid that keeps its error below 1%, against 2.5 s for SMIS at 1% precision (one core); it does not extend to

Table 4 30 routes against the quadrature reference: absolute relative error (median / maximum), relative variance per unit of work (median), and the one-core time for 1% precision at 95% confidence including build and pilot (median, s; measured on a shared machine). Glasserman–Li: factor shift at the dominant point, inner exponential shift, 2×10^4 draws.

	$\gamma = 10^{-4}$			$\gamma = 10^{-6}$		
method	error	RV	time	error	RV	time
SMIS	0.11 / 0.51%	4.6	2.6	0.16 / 0.58%	5.8	3.3
CE mixture	0.16 / 0.38%	4.6	4.6	0.29 / 0.85%	5.9	6.4
single CE shift	0.41 / 23.5%	4.3	2.5	0.36 / 94.7%	5.5	2.5
Glasserman–Li	3.1 / 47.1%	11	5.4	2.9 / 47.6%	13	6.6
conditional Monte Carlo	2.6 / 93.7%	6.5	9 310	3.1 / 63.0%	1.4	2 190
subset simulation	8.1 / 38.1%	716	431	21.6 / 108%	4 430	1 800
routes with $\varrho > 0.75$ only (15): median error						
SMIS	0.15%	4.7		0.15%	5.9	
single CE shift	6.2%	212		9.7%	5.5	
Glasserman–Li	15.5%	12		7.4%	13	

models in which the overflow event couples loads with other random quantities, while the mixture does.

Two end modes suffice. Keeping only the two end components of SMIS (same weights, same defensive component) gives median errors of 0.08–0.19% and relative variances of 4.5–5.9 on the same 30 routes, as accurate as the full mixture: at the levels of interest the interior stages add nothing measurable, in line with Corollary 1.

Longer routes. On eight routes with 15 to 33 stops, against crude Monte Carlo with 5×10^7 draws at 10^{-3} and 10^8 at 10^{-4} (relative standard errors 0.45% and 1.0%), SMIS deviates by at most 0.93% and 1.5%, within two standard errors of the reference. The single shift is accurate on the seven Guangdong routes ($\varrho \leq 0.41$) and fails on the balanced one ($\varrho = 0.96$).

Model variations. On 30 Guangdong routes at 10^{-4}, SMIS stays within its sampling error (median 0.27–0.43% against a relative standard error of 0.47%) for $\rho \in \{0.3, 0.8\}$, coefficient of variation 0.2 and 0.5, pickups scaled by 0.5 and 2, and lognormal marginals, a family the theorems do not cover.

Coverage. Over 300 independent replications of 10^4 draws, the nominal 95% interval of SMIS covers the quadrature value in 94.4% of the replications on 10 Guangdong routes at 10^{-4}, and in 94.9% and 95.3% on 10 balanced Vietnamese routes at 10^{-5} and 10^{-6}. On the balanced routes the single shift, rebuilt in every replication, covers in 34% and 36%: it is biased by -5.2% and -8.2% in median against a replication standard deviation of 13.6%, the picture of Proposition 1 at finite N.

7.5 Several routes

For the 148 depot areas, whose routes share the two factors, the mixture of Corollary 3 has up to 447 components (median 80). Against crude Monte Carlo with 2×10^7 draws on seven areas at 10^{-3} the differences are -0.9 to $+1.6\%$ (standardised -1.2

to 2.1), and against the quadrature program on six of them -0.07 to $+0.29\%$. At the planned capacity no area meets 10^{-3} (the median area overflows with probability 8–11%); to reach 10^{-3} the capacity must grow by 35% (Vietnam) and 43% (Guangdong) in median, and 64% and 76% for 10^{-5}. Treating the routes of an area as independent overstates its overflow probability by a factor of 2.0–2.7 in median: the shared factors matter.

8 Conclusion

The peak load of a pickup-and-delivery route under gamma demands and one-factor Gaussian copulas has logarithmic tail $-c/\lambda^\star$ with an explicit spectral scale, and on that scale it overflows at one of the two ends. This structure decides the behaviour of importance sampling: one shift is efficient only when the weaker end scale is below a quarter of the stronger one; when it is not, the estimator typically misses the weaker mode and reports a precise but low value, with a deficit equal to the mode's share; and a mixture of shifts at the stage dominating points is efficient under a covering condition that can be checked route by route. Three questions are left open: the efficiency of the mixture at computed rather than asymptotic dominating points, which the numerical checks support but we do not prove; continuum mixtures for stages with near ties at large scale; and the case of correlated delivery and pickup factors, in which the two end blocks are no longer independent.

Declarations

- **Funding.** No funding was received for this work.
- **Competing interests.** The author has no competing interests to declare.
- **Ethics approval and consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.
- **Data availability.** The instances, plans and all results used in this paper are in the public repository https://github.com/vinhqdang/stochastic_vrp.
- **Materials availability.** Not applicable.
- **Code availability.** The simulators, estimators and the scripts that produce every number in this paper are in the same repository (directory `svrpspd_wdro`): Table 1 from `scripts/run_rarecert_peis_conditions.py`, Table 3 from `scripts/run_rarecert_peis_bias.py`, and the covering constant of Definition 1 in `rarecert/covering.py`.
- **Author contribution.** The author is responsible for all parts of the work.

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